

Mini-project 3: Solar and Terrestrial Spectra

Purpose

In this mini project you will compare solar and terrestrial spectra with blackbody radiation and use MODTRAN to examine how CO₂ and CH₄ influence the outgoing longwave radiation from Earth. The project links spectral features to atmospheric temperature, emission height and greenhouse-gas absorption.

For each part, first make a brief prediction, then calculate or observe the result, and finally explain whether the result agrees with your prediction. The project is designed for about 8-10 hours of work.

Learning goals

After completing the mini project, you should be able to:

- Compare the solar spectrum with blackbody radiation.
- Calculate the solar constant by integrating a spectral irradiance curve.
- Interpret terrestrial infrared spectra using blackbody curves, brightness temperature.
- Identify important greenhouse-gas bands and explain CO₂ band saturation.
- Use MODTRAN output to quantify changes in outgoing infrared flux when CO₂ and CH₄ are varied.
- Explore the saturation effect
- Explore the concept of emission height
- Present a reproducible analysis with correct units, clear figures and documented code.

Theoretical background

For a blackbody, the spectral radiance per unit wavelength is given by Planck's law,

$$F(\lambda) = \frac{2hc^2}{\lambda^5} \frac{1}{(\exp(hc/k_B\lambda T) - 1)}$$

where $B_\lambda(T)$ is the blackbody spectral radiance, h is Planck's constant, c is the speed of light, k_B is Boltzmann's constant, λ is wavelength, and T is temperature.

The total emitted flux from a blackbody is given by the Stefan–Boltzmann law,

$$F = \sigma T^4,$$

where F is the emitted flux in W m^{-2} , σ is the Stefan–Boltzmann constant, and again T is absolute temperature.

R_{SE} is the radius of Earth's orbit around the sun, and R_{Sun} is the radius of the sun. Then we must multiply by a factor on the equation $1/10^9(3)$ to convert from $\text{W m}^{-2} \text{ m}^{-1}$ to $\text{W m}^{-2} \text{ nm}^{-1}$ so we get a wavelength interval of nm. We then get

$$I_{Sun}(\lambda) = \left(\frac{R_{Sun}}{R_{SE}}\right)^2 \frac{2\pi hc^2}{\lambda^5} \frac{1}{(\exp(hc/k_B\lambda T) - 1)} \quad (1)$$

Here are $I_{Sun}(\lambda)$ Reach the irradiance with units $\text{W m}^{-2} \text{ nm}^{-1}$ which we can compare with extraterrestrial measured spectra. Here you have some parts of the code to help you get started

```
import numpy as np
import matplotlib.pyplot as plt
# Calculate black body radiation intensity W m^-2 nm sr^-1
def BlackBody(w,T):
    # parameters:
    # w Wavelength in meters
```

```

# T Temperature in Kelvin
# Returns:
# Spectral radiance in units W m^-2 nm sr^-1
# Constants
h = 6.62606957e-34 # joule.s
c = 2.99792458e8 # Speed of light in m/s
k = 1.38e-23 # Boltzmann constant in m^2 kg / s^2 / K
return (2*h*c**2/w**5)*(1/(np.exp((h*c)/(w*k*T))- 1))

def BlackBodySun(w,T,R_sun_earth=149600000):
    # Calculate the irradiance from the sun at average sun earth distance.
    # R_sun_earth distance given in km
    # return spectral irradiance W m^-2 nm^-1
    R_sun=696340 #km
    return (R_sun/R_sun_earth)**2*np.pi*BlackBody(w,T)*1.0e-9

def StefanBoltzmann(T):
    return 5.670374419e-8*T**4

```

The sun emits around 5600 to 6000 K The Earth emits mainly in the thermal infrared. When you compare the solar spectrum on top of the atmosphere with the blackbody curve, you will see how well the black body curve as a model fit.

Part 1 — Solar spectrum versus blackbody spectrum

Main idea

Compare the provided Kurucz solar spectrum with a 5778 K blackbody spectrum and estimate the solar constant.

Tasks

1. Load kurudz_0.1nm.dat, check the units and convert the irradiance to $\text{W m}^{-2} \text{nm}^{-1}$.
2. Plot the measured spectrum with a correctly scaled 5778 K blackbody curve.
3. Integrate the measured spectrum over wavelength to estimate the solar constant. (here you can use the trapezoid integration from numpy `np.trapezoid()`)
4. Explain what you observe and try to explain the main differences.

Provided files

kurudz_0.1nm.dat

The file contains spectral irradiance at 1 AU, with wavelength in nm and irradiance in $\text{mW m}^{-2} \text{nm}^{-1}$. (It is distributed with the radiative transfer package libRadtran.)

Files are attached the assignmnet.

Source: <https://www.libradtran.org>

Deliverables for Part 1

1. A figure comparing the Kurucz spectrum with a 5778 K blackbody spectrum. Find a way to visually present the deviation and discuss the best visual way to present the deviations.
2. The calculated solar constant, with units and a short explanation of the method.
3. A brief interpretation of the peak and the main deviations from blackbody radiation.
4. The Python code in an appendix.

Part 2 — Terrestrial infrared spectrum and blackbody curves

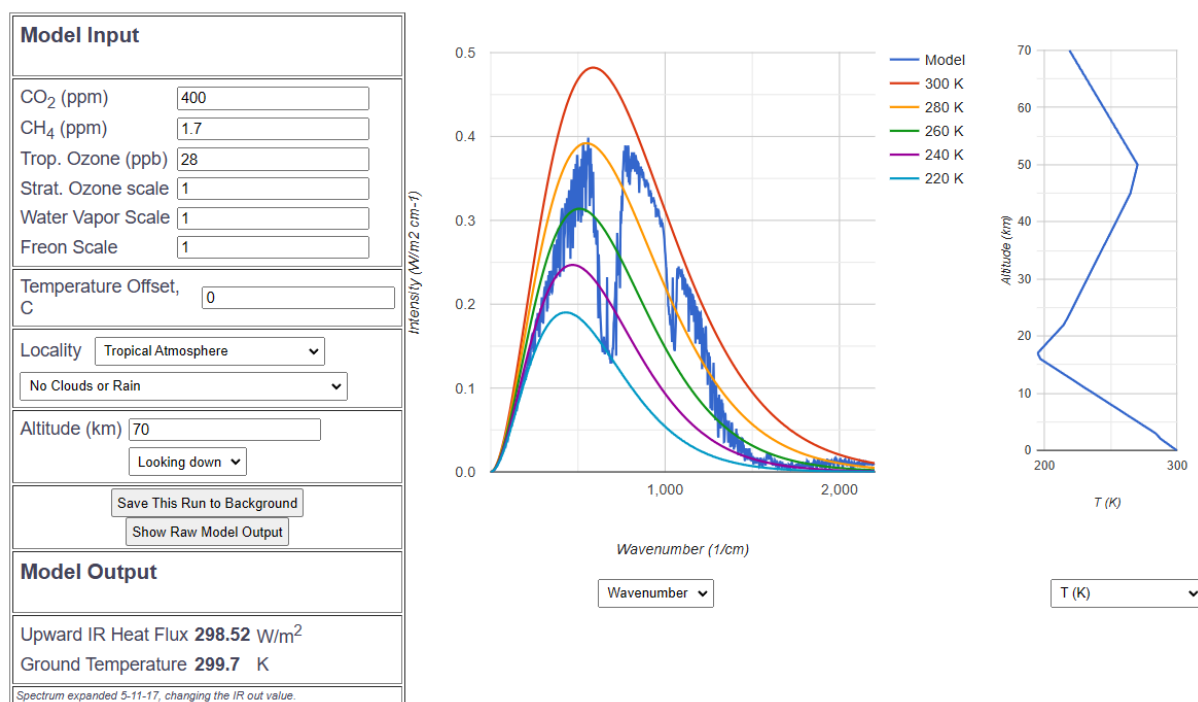
Main idea

Use the UChicago MODTRAN online tool to compare outgoing longwave radiation with blackbody curves and to relate spectral features to atmospheric temperature.

MODTRAN model

<https://climatemodels.uchicago.edu/modtran/>

Open the [MODTRAN model](#) and allow a few seconds for the controls and graphs to load. This model explores how Earth's atmosphere absorbs and emits thermal infrared radiation.



An absolutely minimal tutorial

Choose an atmosphere and viewing position.

Under Locality, select an atmospheric profile, such as *Tropical Atmosphere* or *Midlatitude Summer*, and a cloud condition. Start with “No Clouds or Rain”, Altitude = 70 km, and Looking down to examine radiation escaping toward space.

Adjust the atmospheric inputs.

Control	What to enter
CO ₂ / CH ₄	Gas concentrations in ppm
Trop. Ozone	Tropospheric ozone concentration in ppb
Strat. Ozone scale / Water Vapor Scale	Multipliers: 1 keeps the standard amount; 2 doubles it
Temperature Offset, C	Temperature change relative to the selected profile; start at 0

Here it is useful to be careful and change one parameter at a time. After editing a number, press **Tab** or click outside the field and allow the output to update.

Read the results.

The jagged Model curve is the calculated spectrum; smooth curves show blackbody emission at labelled temperatures. Upward IR Heat Flux, in W m^{-2} , gives the integrated outgoing infrared flux. Use the Wavenumber/Wavelength menu to change the horizontal axis, and the profile menu to inspect temperature or gases versus altitude. [Interpretation guide](#)

Try a CO₂-doubling experiment

Set CO₂ to 400 ppm and record the upward flux. Click Save This Run to Background, change CO₂ to 800 ppm, and compare the new spectrum and flux while holding other inputs fixed. Calculate

$$\Delta F = F_{800} - F_{400}.$$

A negative result means less infrared energy escapes under those conditions. The background overlay makes spectral changes easy to identify.

Explore further.

Repeat with clouds or a different water-vapor scale. To examine infrared radiation reaching the surface from the atmosphere, select Altitude = 0 km and Looking up. Show Raw Model Output opens the numerical model results.

--- End of the minimal tutorial.---

Suggested atmospheric setup

Use the Midlatitude Summer profile, CO₂ = 427 ppm and CH₄ = 1.939 ppm. Leave other settings at their standard values and record all settings used. Relevant data sources for present-day greenhouse-gas concentrations you find from NOAA.

NOAA CO₂ trends: <https://gml.noaa.gov/ccgg/trends/>

NOAA CH₄ trends: https://gml.noaa.gov/ccgg/trends_ch4/

Tasks height sweep

1. Run MODTRAN near 100 km looking down and save the raw spectral output.
2. Repeat the comparison at selected lower heights, for example 80 km, 60 km, 40 km, 20 km and near the surface 0 km.
3. Explain how the spectrum changes with observer height, including the CO₂ band near 15 μm , and check your interpretation against the atmospheric temperature profile.

A Python script that parses and plots MODTRAN output is provided on Blackboard:

Deliverables for Part 2

1. One clear figure showing the spectra plotted for different heights. Add black body curves at 220K, 240K, 260K, 280K, 300K in the background. All wavelength units should be in nm.
2. A short interpretation of the atmospheric window, absorption bands, emission temperatures and height dependence.
3. The Python code used for plotting, should be included in the report appendix.

Part 3 — CO₂ sweep

Objective

Vary CO₂ while keeping all other atmospheric parameters fixed.

Use the same atmospheric profile and model setup as in Part 2 and keep the output height at 100km. In this part, only CO₂ concentration should be changed.

CO₂ values

Run MODTRAN for the following CO₂ concentrations and store the modelled data

0, 1, 10, 50, 100, 200, 300, 400, 425, 500, 600, 700, 800 ppm.

NB! For each run, note the upward infrared heat flux, 'IR out', in W m⁻².

Tasks

1. Run MODTRAN for each CO₂ value and save spectra for a representative selection.
2. Record the total upward infrared heat flux for every run.
3. Plot selected spectra together and plot upward infrared heat flux against CO₂ concentration.
4. Also plot the total outgoing flux against ln(CO₂) and identify the range over which the response is approximately logarithmic.
5. Try to explain why the band centre changes little while the band wings continue to widen.

Use the figures to explain what 'saturation' means and why additional CO₂ can still reduce outgoing longwave radiation.

Deliverables for Part 3

1. A figure that is useful to explain what you have done for comparison CO₂ concentrations.
2. A table and a related graph showing the upward infrared heat flux versus CO₂ concentration.
3. A short explanation of what we mean by saturation, band widening and the approximately logarithmic response.
4. The plotting code in an appendix.

Part 4 — Methane sweep

Vary CH₄ while keeping CO₂ fixed at 427 ppm and all other settings unchanged.

Methane values

Use a physically relevant range of methane concentrations, such as:

0, 0.5, 1.0, 1.5, 1.9, 2.5, 3.0, 4.0, 5.0 ppm.

For each run, record the upward infrared heat flux, "IR out", in W m⁻².

Tasks

1. Run MODTRAN for each methane concentration and save spectra for selected values.
2. Record the total upward infrared heat flux for every run.
3. Plot selected spectra together and plot upward infrared heat flux against CH₄ concentration.
4. Compare the methane response with the CO₂ response from Part 3.
5. Explain the main similarities, differences and limitations of varying methane in isolation.

Your explanation should identify methane's main infrared band and compare both the spectral change and the change in total flux with the CO₂ results.

Deliverables for Part 4

1. A spectral comparison for selected methane concentrations.
2. A graph and table of upward infrared heat flux versus methane concentration.
3. A short comparison of the methane and CO₂ responses.
4. The plotting code in an appendix.

Final report

Each student submits an individual report in Blackboard. Aim for 3-5 pages, plus appendices.

The report must include:

1. A brief introduction and figures with discussion for all four parts.
2. Correct units, axis labels, captions and clearly stated data sources and model settings.
3. Tables of CO₂ and CH₄ concentrations with the corresponding upward infrared heat flux.
4. A concise discussion and conclusion that connect the spectral results to the greenhouse effect.
5. Runnable source code in appendices.
6. The official AI declaration and a short record of how AI was used.

Use of AI

AI tools may be used, but you must document their use and remain responsible for the scientific work.

Appropriate uses include help with plotting code, debugging, error messages, figure design and report structure.

You must understand and be able to explain every figure, calculation, code segment and conclusion in your report.

Include the official AI declaration, examples of important prompts or requests, and a short explanation of how you checked or modified AI-generated output.